

ANKIT TRIPATHI

Backend Developer | Full Stack Engineer (MERN) | AI Systems

Greater Noida, Uttar Pradesh

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Summary

Backend-focused Full Stack Engineer (2025 Graduate) specializing in distributed systems, microservices-based architectures, queue-based processing, and AI-powered applications. Built production-grade AI platforms including a multi-agent repository auditing system, an AI-powered infrastructure deployment platform with on-chain payments, and an intelligent LLM orchestration engine with validated presentation generation.

Education

Galgotias University

Bachelor of Technology in Computer Science & Engineering (CGPA: 7.48)

2022 – 2025

Greater Noida, UP

Relevant Coursework

Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Object Oriented Programming, System Design

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Java, SQL

Backend: Node.js, Express.js, Fastify, REST APIs, Microservices Architecture, BullMQ, Socket.IO, JWT, OAuth 2.0, Zod, Prisma ORM

Frontend: React.js, Next.js, Redux Toolkit, TailwindCSS

Databases: PostgreSQL, MongoDB, Redis

AI & Automation: LLM Integrations, Multi-Agent Systems, Prompt Engineering, AI Workflow Orchestration

DevOps & Tools: Docker, Git/GitHub, GitHub Actions (learning), Postman, Linux, Railway, Vercel, AWS Basics

Core Concepts: System Design, Queue-Based Processing, Caching Strategies, Real-time Systems, ACID Transactions, JWT Auth, RBAC, OOP Design Principles

Projects

AI Software Engineering Team | [Live Demo](#) | [GitHub](#)

2026

- Built an AI-powered repository auditing platform using a multi-agent architecture with 5 specialized agents for architecture, security, performance, dependency, and code-quality analysis.
- Architected the system as a set of decoupled microservices — each AI agent runs as an independent processing unit communicating via Redis queues, enabling parallel execution and independent scaling.
- Designed asynchronous job processing with BullMQ and Redis to handle long-running repository scans without blocking, with dead-letter queues and retry-with-backoff for fault tolerance.
- Implemented real-time scan progress updates using Socket.IO, eliminating polling and reducing client-side latency.
- Integrated PostgreSQL with Prisma ORM for structured report storage, scan history, pagination, and audit retrieval.
- Generated automated PDF audit reports and built a Next.js dashboard with JWT auth, refresh tokens, and scan submission UI.

Neural Nodes – AI Infrastructure Deployment Platform | [GitHub](#)

2026

- Built an AI-powered deployment platform where users paste a GitHub repository and the system automatically analyzes architecture, detects stack (frontend/backend/fullstack), calculates pricing, and provisions a live URL.
- Engineered an AI-based repository analysis layer that classifies project structure and generates dynamic infrastructure cost estimates without manual configuration.
- Integrated Locus Checkout for USDC on-chain payments with webhook-based deployment automation — payment confirmation triggers deployment pipeline automatically.
- Designed real-time deployment status flow using webhooks and event-driven architecture, enabling both human users and AI agents to autonomously pay and execute infrastructure tasks.

Code EZ – AI-Powered Markdown to PPTX Engine | [GitHub](#)

2026

- Built an intelligent Markdown-to-PowerPoint generation pipeline using Groq (Llama-3.3-70b) as a structured data extraction agent — not a simple LLM wrapper.
- Engineered a custom validation guardrail (`engineLogic.js`) that intercepts LLM chart suggestions, mathematically verifies data suitability (e.g., pie chart requires values summing to ~100%), and auto-corrects invalid choices — achieving 100% fault tolerance.

- Designed a dynamic layout engine with adaptive negative-space optimization and smart overflow protection via Y-axis shifting for pixel-perfect rendering across all content densities.
- Achieved 100% statement coverage on the chart normalization engine using Jest and Supertest, handling edge cases including negative values, empty arrays, and string-number mismatches.

BookIt – High-Concurrency Booking Engine | [Live Demo](#) | [GitHub](#)

2025

- Engineered a collision-free slot booking algorithm using MongoDB transactions to prevent double-booking under concurrent load.
- Implemented Redis caching for frequently accessed experience slots, reducing database load by 40%.
- Integrated Stripe API for secure payment processing and Zod for strict schema validation.

Key Achievements

- Designed a fault-tolerant async job pipeline handling parallel AI agent execution with dead-letter queues and retry-with-backoff for production resilience.
- Built an end-to-end AI deployment platform where webhook-verified on-chain payments automatically trigger infrastructure provisioning — no manual intervention.
- Achieved 100% statement coverage on an LLM chart normalization engine using Jest, handling edge cases including negative values, empty arrays, and type mismatches.
- Reduced database load by 40% in a high-concurrency booking system using Redis caching and query optimization.